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Activity #1 · Variables

Part A · Variable Identification Table

Variable name	Data type	Variable type	Stored in
name	String	Instance Variable	Heap Memory
grade	int	Instance Variable	Heap Memory
n	String	Parameter	Stack Memory
g	int	Parameter	Stack Memory
args	String[]	Parameter	Stack Memory
S1	Student	Reference Variable	Stack Memory
S2	Student	Reference Variable	Stack Memory
S3	Student	Reference Variable	Stack Memory

Part B: Explanation, Please see example below.

Statement	What happens?	Memory Impact
Student s1 = newStudent("John",85);	A new Student object is created with name="John" and grade=85. Reference variable s1 points to this object	Stack: s1 reference created. Heap: new Student object allocated with name="John", grade=85
Student S2 = newStudent("Maria",90);	A new Student Object is created with name="Maria" and grade=90. Reference Variable S2 points to this object.	Stack: S2 reference created Heap: new Student object allocated with name="Maria", grade=90
Student S3=S2;	No new object is created. S3 copies the reference from S2, so both S3 and S2 point to the same object	Stack: S3 reference created Heap: no new object, S3 & S2 now share the same memory address.

Statement	What happens?	Memory impact
S1.grade = 95;	The grade field of the object reference by S1 is updated from 85 to 95.	Heap: S1's object is modified, grade changes from 85 to 95. No Stack change.
S3.name = "Ana"	The name field of the object is referenced by S3 is updated. Since S3 and S2 shares the same object S2.name also becomes "Ana"	Heap: The shared object (S2 and S3) is modified, name changes from "Maria" to "Ana". No Stack change.

Part c - Output

"C:\Program Files\Java\jdk-25\bin\java.exe" "-javaagent:C:\Program Files\Jetbrains\IntelliJ IDEA 2025.3.3\lib\idea_rt.jar=54289" -DFile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8 -classpath C:\Users\lanzel\IdeaProjects\Variables\out\production\Variables TestStudent

John 95

Ana 90

Ana 90

Process finished with exit code 0